

CSE 4417 IoT Theory

Lecture 07: Networking Part 2

Atanu Shuvam Roy



Lecturer,
Dept. of CSE, ULAB, Dhaka



EMAIL:

atanu.shuvam@ulab.edu.bd



Class Code
bjedchaj



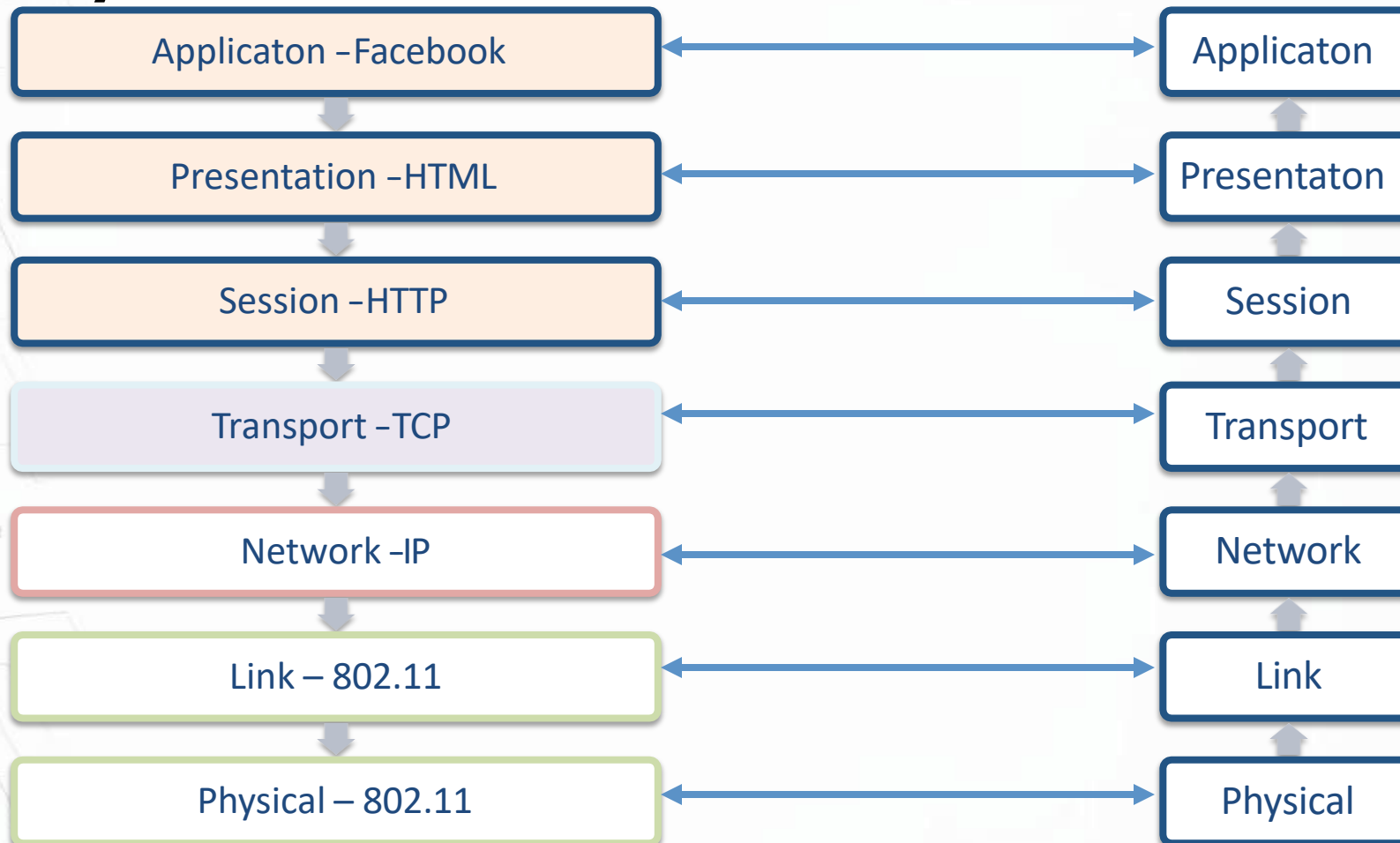
Spring 2026

Internet of Things

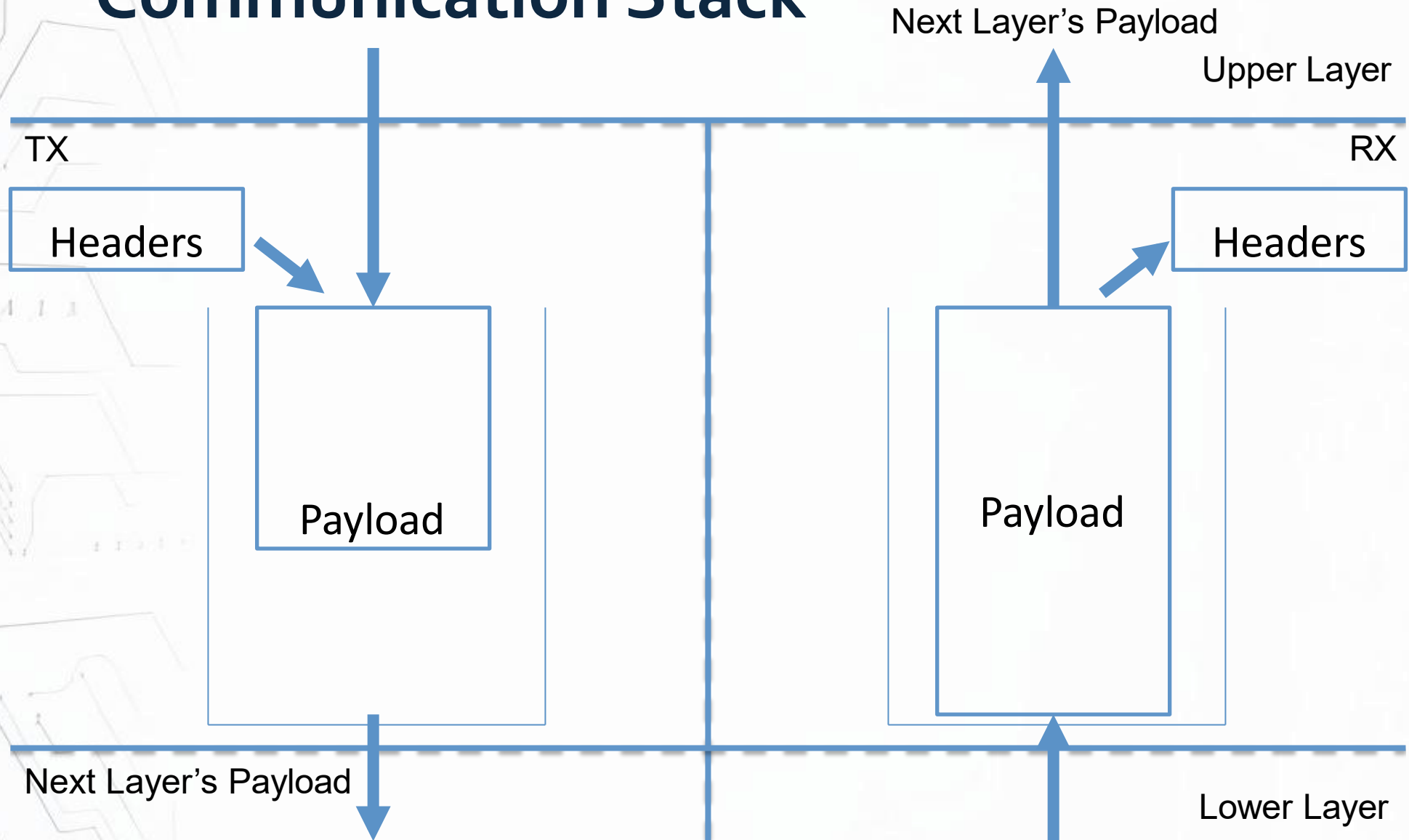
- **Networking**
 - link (machines, especially computers) to operate interactively
- **Communication**
 - the imparting or exchanging of information or news



OSI Layers



Communication Stack



The connection problem in IoT

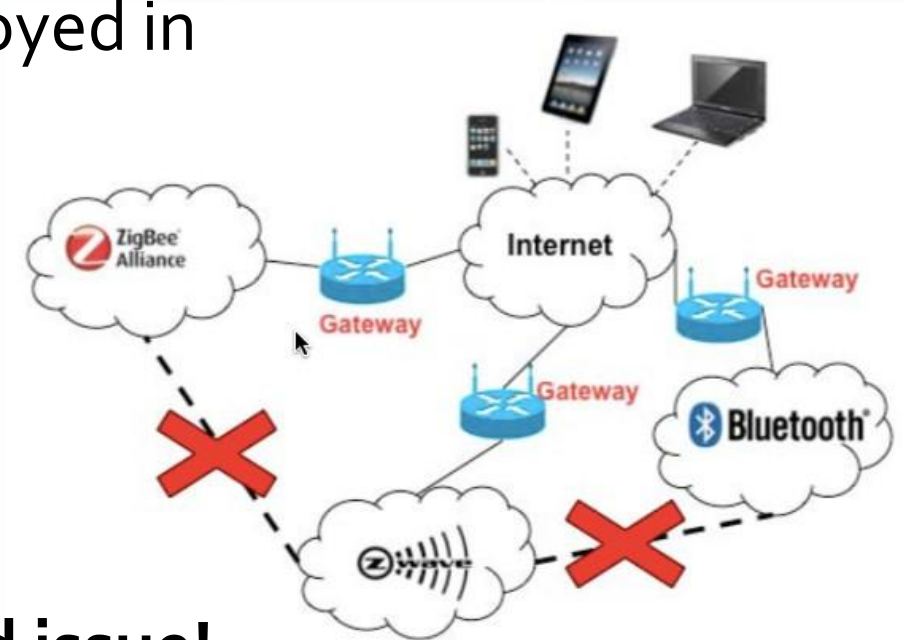
Many different protocols are deployed in WSNs

They cannot understand each other

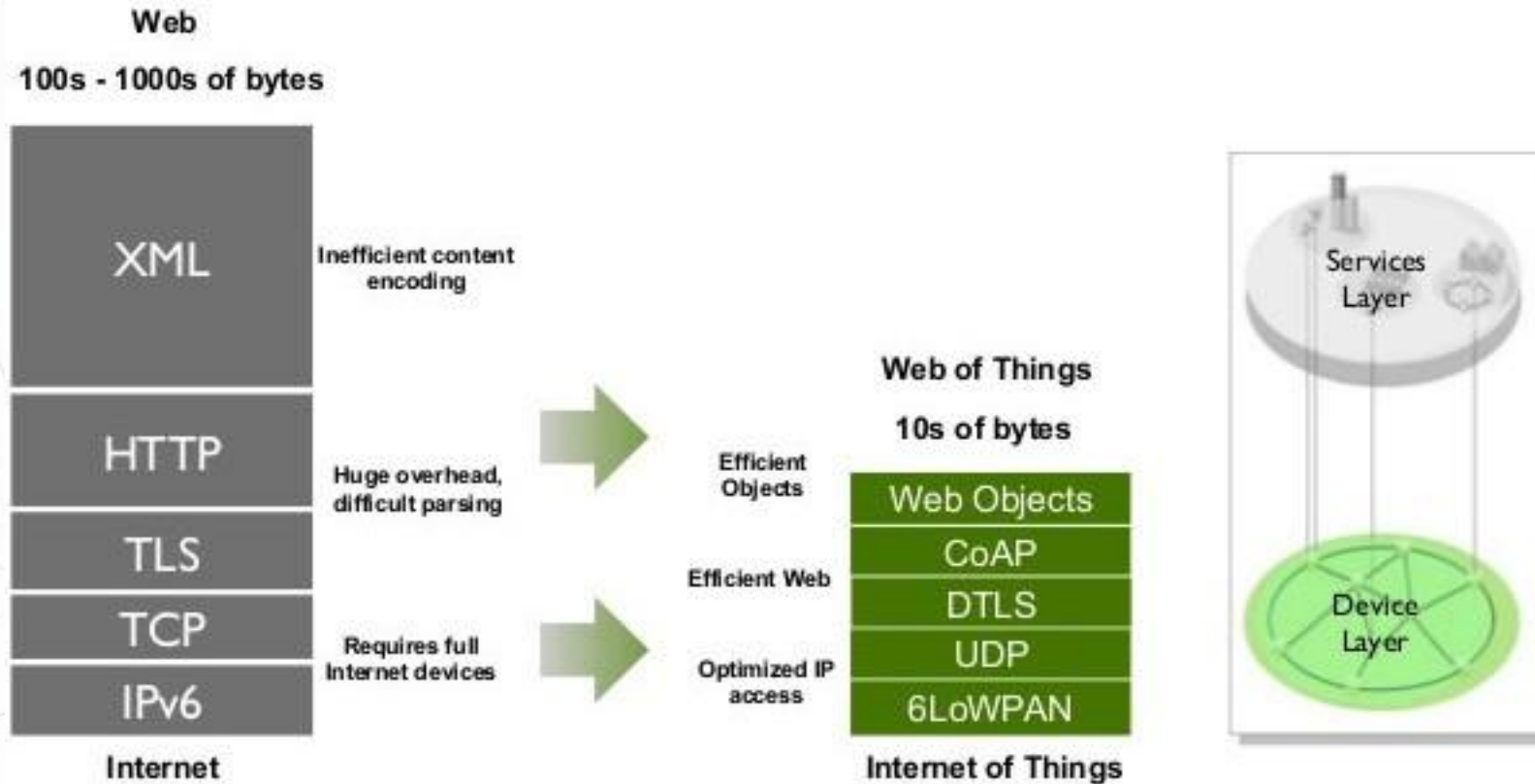
and this is **NOT** a software related issue!

Common solution:

gateways connecting WSNs to internet



Current internet is not enough

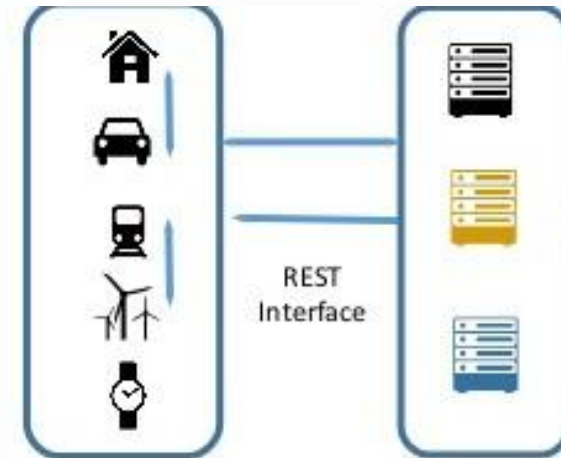


Application Layer

- Commonly designed with Presentation and Session
- Common data representation and data retrieval
- Lighting automation, home automation, distributed control, Skype, Facebook, Hangouts
- Mostly unaware of underlying infrastructure

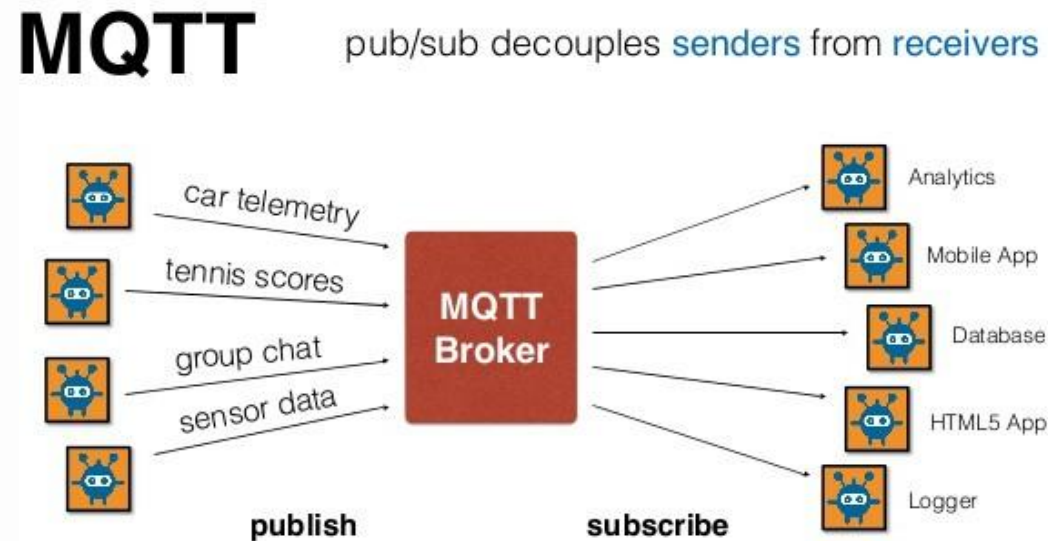
Application Layer in IoT

- WSNs with embedded devices are memory, CPU, energy constrained
- CoAP: Constrained Application Protocol
 - Easily convertible to HTTP
 - Supports multicast
 - Very low overhead (4-byte header + TLVs)
 - Data needs to be polled (Put/Get)



Application Layer in IoT

- MQTT
 - Publisher – Subscriber model
 - Light weight, minimizes code on remote end
 - Data is published once available
 - Good for M2M Networks, distributed control apps



Open Problems

- Unification/Standardization:
 - Unified data representation
 - Unified and seamless data translation
- Very small code space and memory requirements
 - Where to store/cache data

Transport Layer

- Segmentation and reassembly
- End-to-end communication reliability
- Congestion control
- Reordering
- Security is added through Transport Layer Security

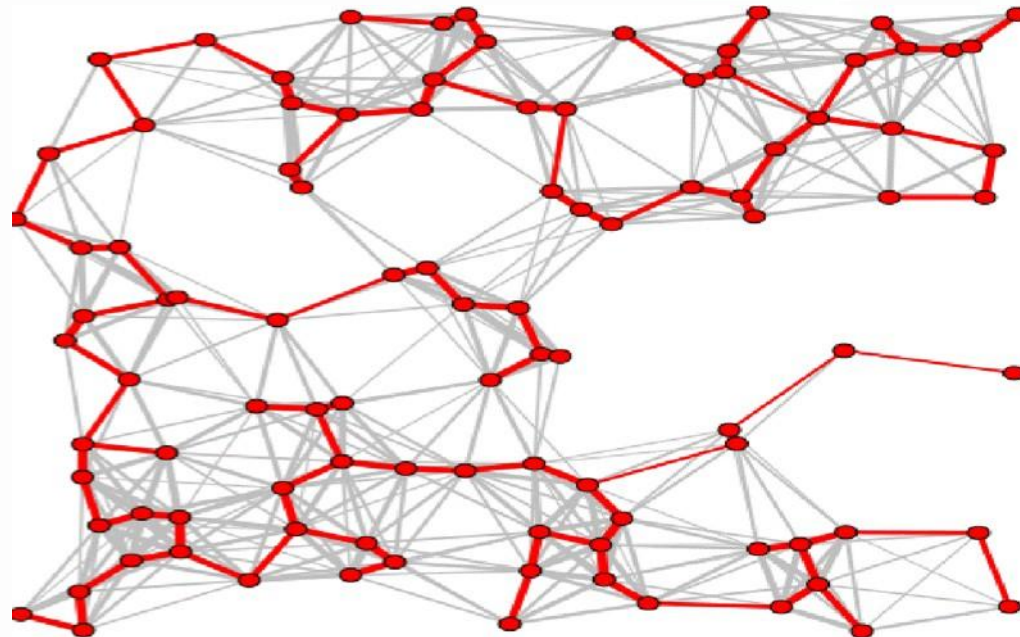
	Seg. & Reass.	End2End Reliability	Cong. Control	Reordering
TCP	YES	YES	YES	YES
UDP	NO	NO	NO	NO
DCCP	NO	NO	YES	NO
SCTP	NO	YES	YES	YES (opt)

Open Problems

- TCP is \$\$\$\$\$:
 - High memory usage
 - End to end communication resource usage
 - High loss links
- UDP is:
 - Not reliable
 - Assumes ordered delivery

Network Layer

- Management of multiple nodes
- Addressing & Routing & Security (IPSec)
- Internet Protocol (IP) is the dominant solution
- There are also other protocols within “suite” solutions

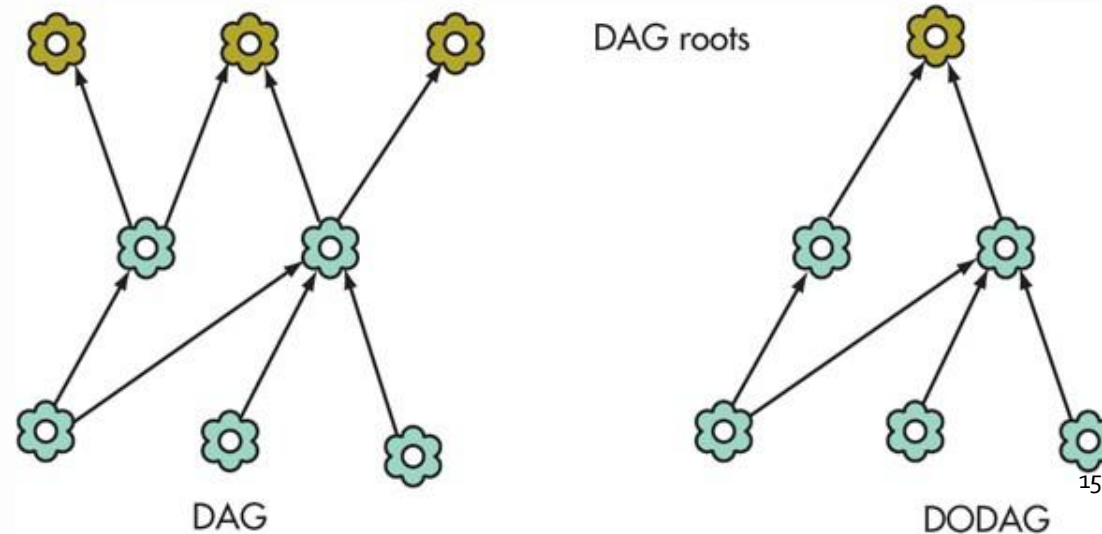


Network Layer in IoT

- Routing Protocol for Low-Power and Lossy Networks (RPL):
 - Dominant routing protocol for IPv6 on WSNs
 - Creates directed trees
 - Flexible architecture for energy, delay, link quality based routing selections
- IP is very inefficient by itself
 - 6LowPAN (IPv6 over Low-Power WPAN) is used for improvement (cross-layer compressing)

RPL

- Creates destination oriented directed acyclic graphs
- Optimized for upward (sink destined) traffic
- Supports mobility (treated as lossy link)
- Supports multiple DODAGs (sinks)
- Supports different routing modes (non/storing mode)

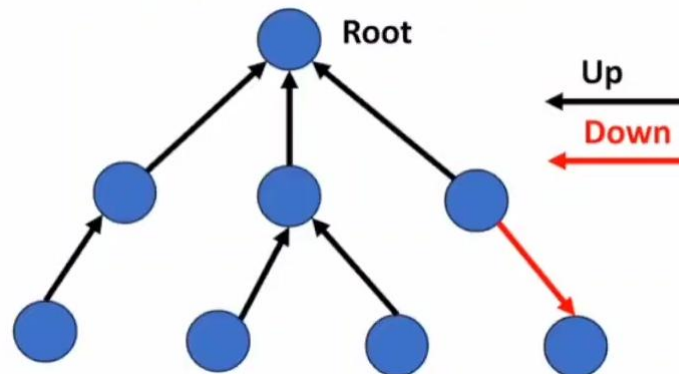


RPL Protocol

- For Low Power and Lossy Networks
- Lossy Networks = High Packet drop rate
- Mainly used with IPv6 but can be used with IPv4
- Distance Vector Routing Protocol
- It is based on Distance Oriented Directed Acyclic Graphs (DODAG)

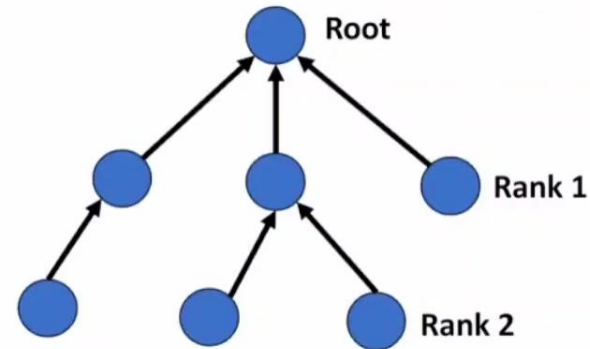
RPL Terminologies

- Directed Acyclic Graph (DAG) – It is a graph that contains no cycle. E.g. Spanning trees
- Root – Destination node of DAG. No outgoing edge.
- Up – Edge towards the root
- Down – Edge away from the root
- DODAG – Special DAG where every node wants to go toward a single destination



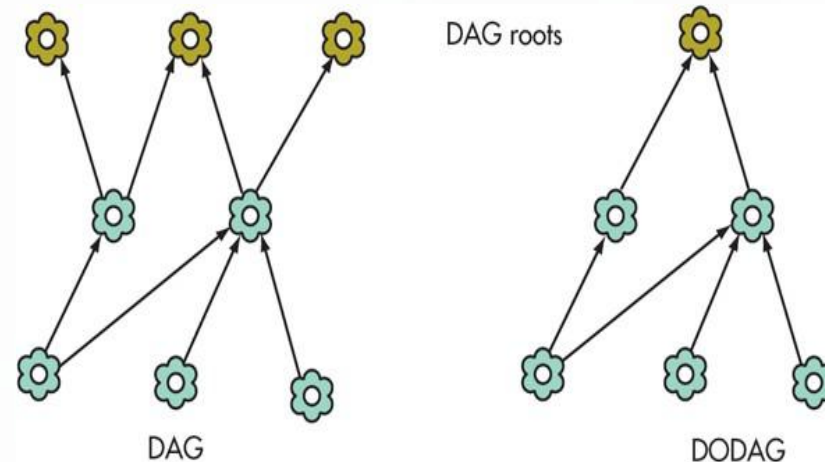
RPL Terminologies

- DODAG – Special DAG where every node wants to go toward a single destination
- Objective function – helps us to decide whether we are near to the root or away from it. And its decided by programmer or designer. We want to minimize transmission, distance of transmission and number of hops.
- Rank – Number of hops



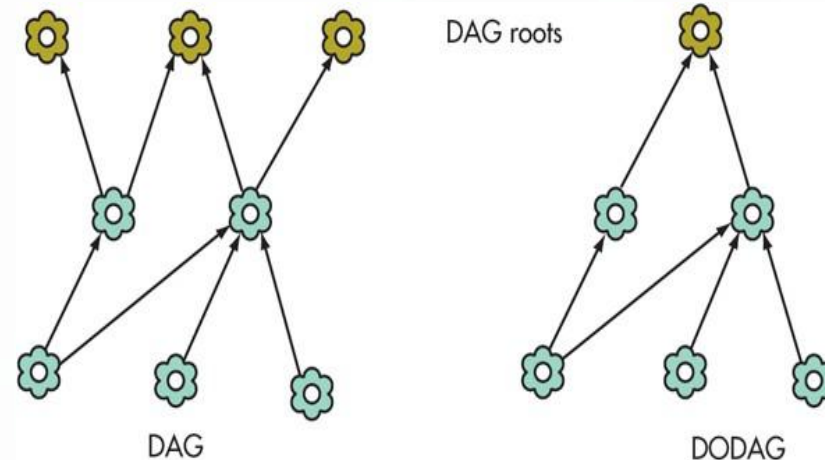
RPL Terminologies

- RPL Instance – When we have one or more DODAGs then each DODAG is an instance
- DODAG ID - Each DODAG has IPv6 ID of 128 bits. This ID is only given to root, and it stays until the route doesn't change
- DODAG Version – New shape of DODAG = new version



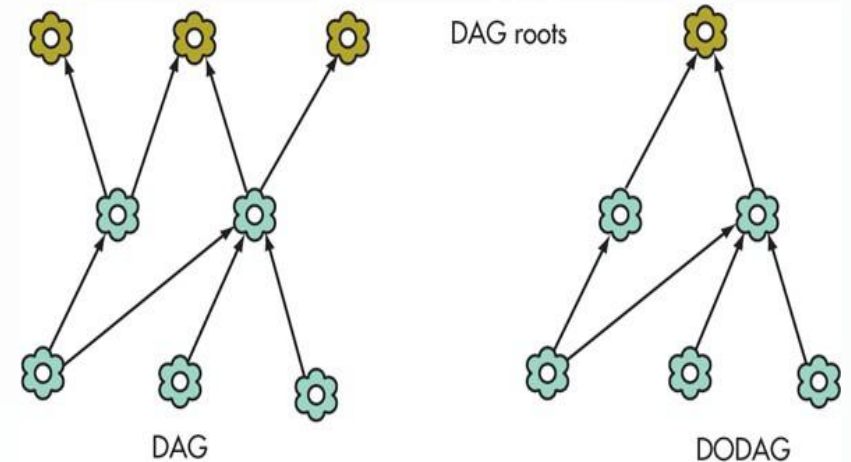
RPL Terminologies

- RPL Goal – Best route using DODAG and aim is to minimize the objective function
- Parent Child Node – Parent = where I am going, child = from where I am coming
- Sub-DODAG – Subtree of given DODAG



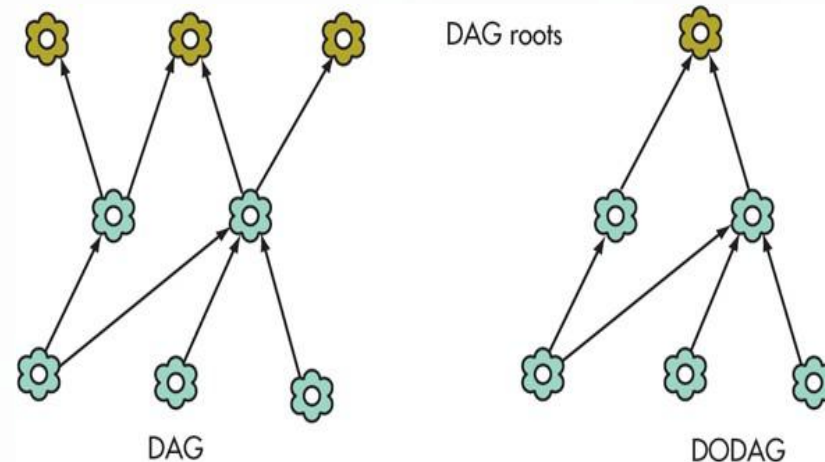
RPL Terminologies

- Storing Nodes – Keeps the whole routing table. Knows how to route the packets from one node to another. Root node is always storing node
- Non-Storing Nodes – Nodes knows about their parents only. They can not restore the routing table
- Grounded – Network is established
- Floating – Network is not established



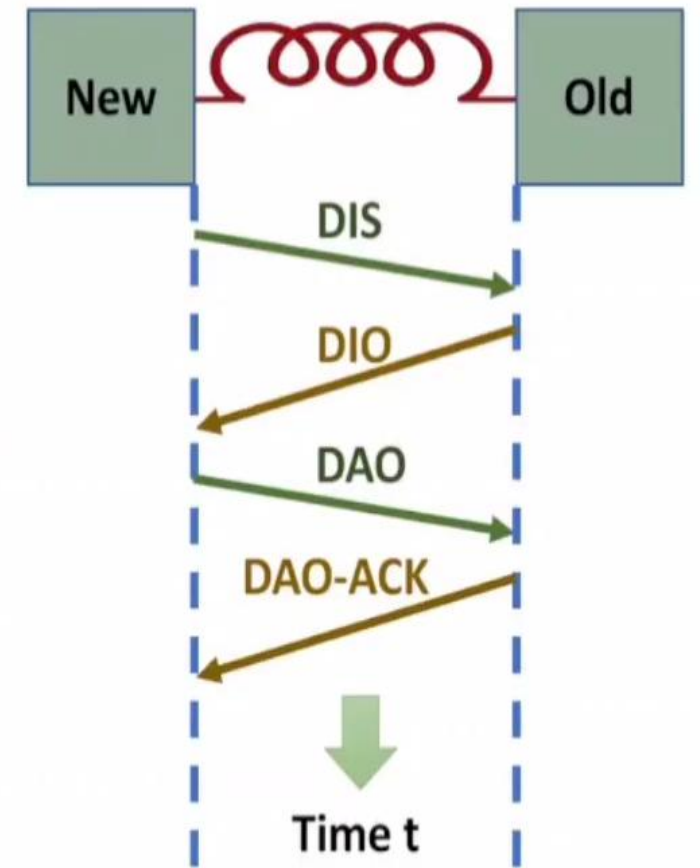
RPL Control Messages

- DODAG Information Solicitation (DIS) – When no announcement is heard, announcement message from new to old node to find if any DODAG exists or not
- DODAG Information Object (DIO) – Multi-casted downwards. This message lets others know, if node is grounded or not, whether it is storing or not, or if they are interested to join the network or not.



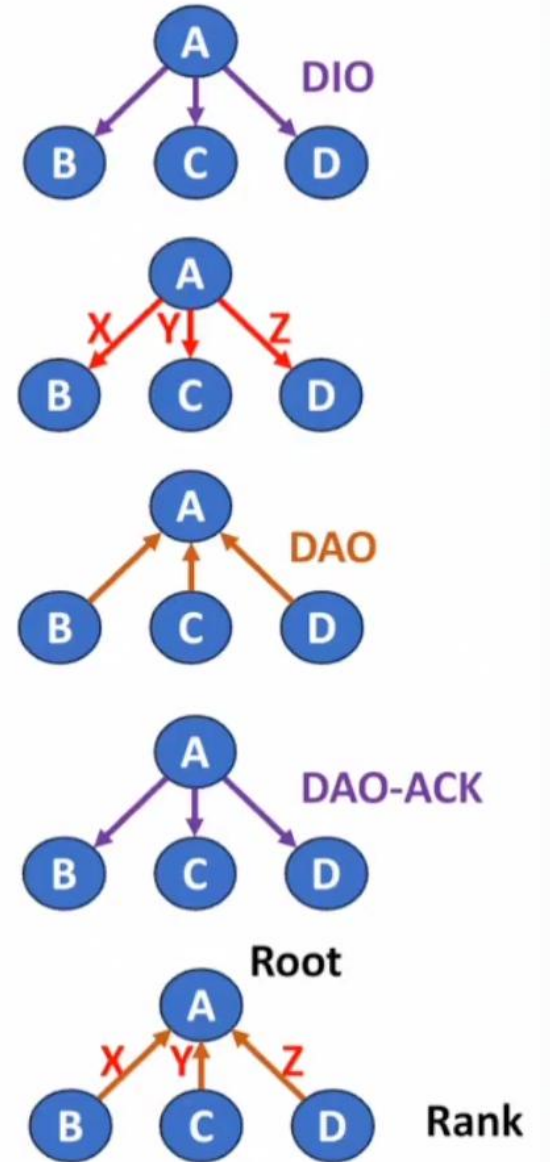
RPL Control Messages

- DODAG Advertisement Object (DAO) – It is a request sent by child to root. This message is a request to allow the child node to join to a DODAG.
- DAO-ACK - Old node to new node -> Response YES or NO
- Consistency Check (CC) – It deals with security for consistency in DODAG



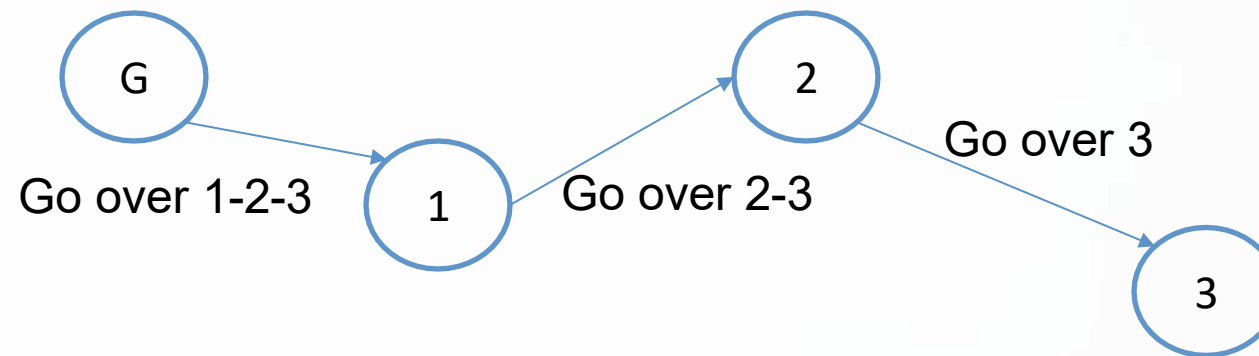
DODAG Formation

- Root in DODAG is special node. Because it has ID.
- Suppose we have 4 nodes: A, B, C, and D. To form DODAG, the following steps are taken:
- Node A (Root) Multicasts DIOs.
- Upon receiving DIOs by rest of the nodes (B, C, D), rest of the nodes will have idea of distance with respect to node.
- Then B, C, D will send DAOs to root node A.
- Root node A accepts DAOs and gives back DAO-ACK to rest.
- B is nearest to A. B will do the same to other nodes. Hence distance with each other will be known and rank will be decided and routing will be decided.



RPL + Memory Problem = Nonstoring Mode

- A routing table entity contains Node ID + Next Hop = 32 bytes per destination
- RAM is a luxury in small devices
- Nonstoring mode solves the memory problem by storing the routing table only at the gateway



Resources to help you

- <https://www.youtube.com/watch?v=UGDHnjJxukI>
- <https://www.youtube.com/watch?v=kSiUGeUgJYQ>

References

- <https://qiriro.com/ecs6264/lectures/>
- <https://cseweb.ucsd.edu/classes/wi16/cse291-c/>
- https://profile.iiita.ac.in/bibhas.ghoshal/IoT_2021/Slides/Application_Protocols.pdf
- https://profile.iiita.ac.in/bibhas.ghoshal/teaching_iot_new.html

Assignment

- Download the following papers and write a 500–1000 word assignment regarding your understanding about the paper.
- Old Papers (10 Years+ ago)
- Paper 1: <https://arxiv.org/abs/1406.6608>
- Paper 2: <https://ieeexplore.ieee.org/abstract/document/6704479/>
- Paper 3: <https://ieeexplore.ieee.org/abstract/document/6064380>
- Recent Review Paper:
Paper 4: <https://www.sciencedirect.com/science/article/pii/S2214785321036439#bi005>
- Do a contrastive analysis on the papers.

You work will be passed through Turnitin. Any AI Detection (>20%) or plagiarism (>10%) writing or paraphrasing will generate a direct 0.



ULAB

UNIVERSITY OF LIBERAL ARTS
BANGLADESH

Thank You

atanu.Shuvam@ulab.edu.bd

University of Liberal Arts Bangladesh
Department of Computer Science & Engineering